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CSP-Activity

Part - 1:

1.1: Problem Foundation:

Variables: C_1, C_2, C_3, C_4, C_5

Domains:

$D_1: \{C\}$

$D_3: \{A, B, C\}$

$D_5: \{B, C\}$

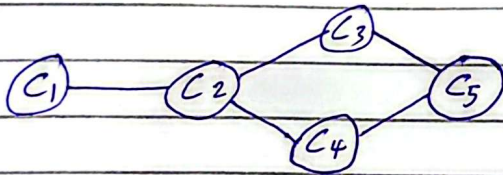
$D_2: \{B, C\}$

$D_4: \{A, B, C\}$

Binary Constraints:

$C_1 \neq C_2, C_2 \neq C_3, C_2 \neq C_4, C_3 \neq C_4, C_3 \neq C_5, C_4 \neq C_5$

1.2. Constraint Graph:



1.3. Arc Consistency:

	Arcs:	Agendas:
$C_1 = \{C\}$	$C_1 \neq C_2$	$C_1 \neq C_2, C_2 \neq C_1, C_2 \neq C_3, C_3 \neq C_2, C_2 \neq C_4,$
$C_2 = \{B, C\}$	$C_2 \neq C_1$	$C_4 \neq C_2, C_3 \neq C_4, C_4 \neq C_5, C_3 \neq C_5, C_5 \neq C_3,$
$C_3 = \{A, B, C\}$	$C_2 \neq C_3$	$C_4 \neq C_5, C_5 \neq C_4, C_1 \neq C_2, C_2 \neq C_3, C_2 \neq C_4$
$C_4 = \{A, B, C\}$	$C_3 \neq C_2$	
$C_5 = \{B, C\}$	$C_2 \neq C_4$	
	$C_4 \neq C_2$	
	$C_3 \neq C_4$	
	$C_4 \neq C_3$	
	$C_3 \neq C_5$	
	$C_5 \neq C_3$	
	$C_4 \neq C_5$	
	$C_5 \neq C_4$	

Arc consistent sol:

$C_1 = \{C\}$

$C_2 = \{B\}$

$C_3 = \{A, C\}$

$C_4 = \{A, C\}$

$C_5 = \{B, C\}$

Ans.

1.4. Solution:

$$C_1 = C$$

$$C_2 = B$$

$$C_3 = A$$

$$C_4 = C$$

$$C_5 = B$$

Part-2:

1.

If class 4 were only taught by professor A, its domain would be $C_4 = \{A\}$.

Then during arc-consistency check, $C_3 = \{B\}$ will be removed due to $C_3 \neq C_2$.

Then $C_3 = \{A, C\}$ left so $C_3 = \{A\}$ will also be removed due to $C_3 \neq C_4$ & $C_3 = \{C\}$ will be left ~~only~~.

2.

Forward checking would be less efficient overall compared to Arc-Consistency for large scale problem of 500 classes. Because forward checking only ~~one~~ looks one step ahead, which misses ~~the~~ deeper conflicts & leads to massive, time-consuming backtracking. Arc-consistency repeatedly enforces local constraints, ~~and~~ catching inevitable failures much earlier.

Part - 3:

2.1.

C R A C K	$\Rightarrow K + K = R + 10c_1$	$\Rightarrow K = 1$	$E = 5$
+ H A C K	$C + C + c_1 = 0 + 10c_2$	$C = 4$	$O = 8$
E R R O R	$A + A + c_2 = R + 10c_3$	$A = 6$	
⁰ 4 ⁰ 2 8 4 1	$R + H + c_3 = R + 10c_4$	$R = 2$	
+ 9 8 4 1	$C + c_4 = E$	$H = 9$	
5 2 2 8 2			

2.2.

B L A C K	$\Rightarrow K + N = E + 10c_1$	$\Rightarrow B = 7$	$R = 3$
+ G R E E N	$C + E + c_1 = G + 10c_2$	$L = 9$	$E = 4$
O R A N G E	$A + E + c_2 = N + 10c_3$	$A = 2$	$N = 6$
⁰ 7 9 2 ⁰ 0 8	$L + R + c_3 = A + 10c_4$	$C = 0$	$O = 1$
⁰ + 5 3 4 4 6	$B + G + c_4 = R + 10c_5$	$K = 8$	
1 3 2 6 5 4	$c_5 = 0$	$G = 5$	

2.3.

E A T	$\Rightarrow E = 8$
+ T H A T	$A = 1$
A P P L E	$T = 9$
8 ⁰ 1 9	$H = 2$
⁰ + ⁰ 9 2 1 9	$P = 0$
1 0 0 3 8	$L = 3$

2.4.

FOUR	$\Rightarrow F=1$	$E=4$
+ FIVE	$O=9$	$N=3$
NINE	$U=5$	
$\begin{array}{r} 010950 \\ + 1284 \\ \hline 3234 \end{array}$	$R=0$	
	$I=2$	
	$V=8$	

2.5.

CROSS	$C=9$	$G=7$
+ ROAD	$R=8$	$E=4$
DANGER	$O=2$	$N=8$
$\begin{array}{r} 96233 \\ + 62513 \\ \hline 158748 \end{array}$	$S=3$	
	$A=5$	
	$D=1$	